

1.

a) What do you mean by true color representation? Does it represent any disadvantages? How to handle the problems?

True Color Representation refers to a way of displaying colors in images using 24 bits.

This allows for 16,777,216 different colors, combining:

- ✓ 8 bits for Red
- ✓ 8 bits for Green
- ✓ 8 bits for Blue

Each color channel can have values from 0 to 255, giving a wide range of colors.

Disadvantages of True Color Representation

- ✓ **Memory Usage:** True color images take up a lot of space. For example, a simple image of 1920x1080 pixels would require about 6 MB (24 bits per pixel).
- ✓ **Processing Speed:** More colors can slow down image processing, especially in applications like gaming or video editing.
- ✓ **Hardware Requirements:** Not all devices can display true color images well, especially older ones.

Handling the Problems

- ✓ **Compression:** Use image formats like JPEG or PNG that compress files without losing too much quality. This helps save space.
- ✓ **Reduce Resolution:** Use smaller image sizes where high detail isn't needed. For example, a 640x480 image requires less memory than a 1920x1080 image.
- ✓ **Color Depth Reduction:** Instead of using true color, sometimes use fewer bits (like 16 bits) for applications where exact colors aren't critical. This can save memory.

b) If we use 5 bits each for red and blue and 6 bits for green in direct coding method, how many possible simultaneous colors do we have?

In the direct coding method, the number of possible simultaneous colors is determined by the number of bits assigned to each color channel.

If we use:

- 5 bits for red,
- 6 bits for green, and
- 5 bits for blue,

The total number of possible colors can be calculated as:

$$2^{\text{bits for red}} \times 2^{\text{bits for green}} \times 2^{\text{bits for blue}}$$

Substituting the given values:

$$2^5 \times 2^6 \times 2^5 = 32 \times 64 \times 32 = 65,536. \text{ So, the total number of possible simultaneous colors is } 65,536.$$

c) Interpret EFE_{16} as unsigned, signed, and floating point numbers that a microprocessor can understand.

To interpret the hexadecimal number EFE_{16} in different numerical formats (unsigned, signed, and floating point), we first need to convert it to binary.

Step 1: Convert EFE_{16} to Binary

The hexadecimal number EFE is equivalent to:

$E = 14$ (in decimal) = 1110 (in binary)

$F = 15$ (in decimal) = 1111 (in binary)

$E = 14$ (in decimal) = 1110 (in binary)

So, EFE in binary is: 1110 1111 1110

The binary representation is 111011111110.

Step 2: Interpret as Unsigned Integer An unsigned integer treats the binary number as a positive value only.

Binary: 111011111110

Decimal: $2048 + 1024 + 512 + 0 + 128 + 64 + 32 + 16 + 8 + 4 + 0 + 0$
 $= 3838$

Step 3: Interpret as Signed Integer

A signed integer uses the most significant bit (MSB) as the sign bit (0 for positive, 1 for negative).

Binary: 111011111110 (MSB is 1, so it's negative)

Two's Complement:

Invert the bits: 000100000001

Add 1: $000100000001 + 1 = 000100000010$

Convert to decimal $= 2 + 1024 = 1026$

Since it was negative, the signed integer is -1026.

Step 4: Interpret as Floating-Point Number

For floating-point representation, we'll assume IEEE 754 single-precision format (32 bits).

The given number is 12 bits long, so we need to pad it with zeros to fit into 32 bits.

Binary Representation: 00000000 00000000 11101111 1110 (add leading zeros to make 32 bits)

IEEE 754 Format

Sign bit: 0 (since it's positive)

Exponent: Next 8 bits (the next 8 bits after the sign bit): 00000000 $\rightarrow$ exponent = 0.

Mantissa (Fraction): Last 23 bits: 000000111011111110 $\rightarrow$ The binary fraction =

1.111011111110

1.111011111110

Convert Mantissa to Decimal

$1+0.5+0.25+0.125+0+0.03125+0.015625+0.0078125+0.00390625+0.001953125+0.0009765625 = 1.9375$ (approximately)

Final Calculation

The exponent bias is 127, so we get: $2^{(0-127)} = 2^{-127}$

Result

Unsigned Integer: 3838

Signed Integer: -1026

Floating Point: Approximately 1.9375×2^{-127} (which is a very small number)

d) What are the common problems with scanner? Share some tips to troubleshoot the basic 3 scanner related problems.

1. Scanner Not Detected

Fixes:

- ✓ Check the cables and power.
- ✓ Try a different USB port or network connection.
- ✓ Update or reinstall scanner drivers.

2. Poor Scan Quality

Fixes:

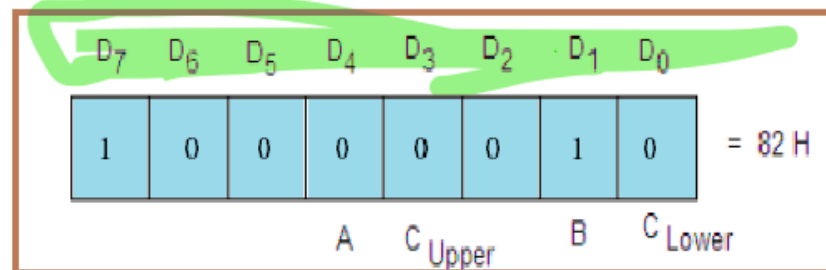
- ✓ Clean the scanner glass.
- ✓ Adjust scan resolution (300 DPI for text, 600 DPI for images).
- ✓ Check for software updates.

3. Scanner Freezes or Slow Scans

Fixes:

- ✓ Close other programs on your computer.
- ✓ Lower the scan resolution.
- ✓ Restart the scanner to clear its memory.

e) Obtain the control word when the ports of 8255A are to be used in mode 0 with port-A as output port and port B as input port and port C as output port.



d) Make a comparison between buffers and spoolers.

- ✓ **Buffer v's Spooler:** If the CPU is very busy and doesn't get much idle time then spooling can be a very slow process.
- ✓ **Spooler v's Buffer:** A buffer is limited by the amount of RAM whereas a spooler uses backing storage which has a very large capacity.
- ✓ Use BOTH for optimum efficiency.

3.

A. Suppose, a processor is having 16 bits address line and it is byte addressable. Interface 4096x8 RAM with the processor.

a) What is the memory capacity of the processor?

The number of address line is 16 bits.

Hence, the memory capacity of the processor = $2^{16} = 64k$

b) How many address lines are required for a RAM?

Addresses are required for a RAM = $2^{12} = 4k$

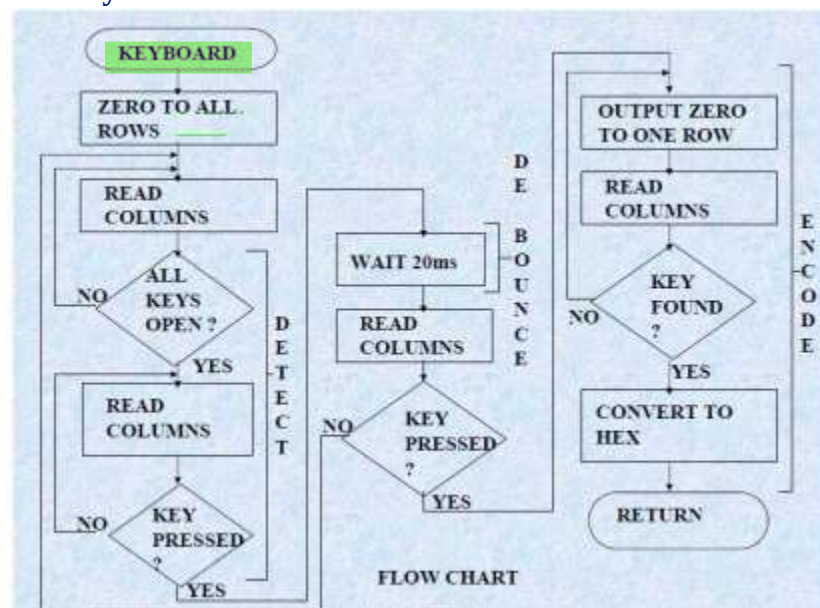
c) How many RAM chips are required?

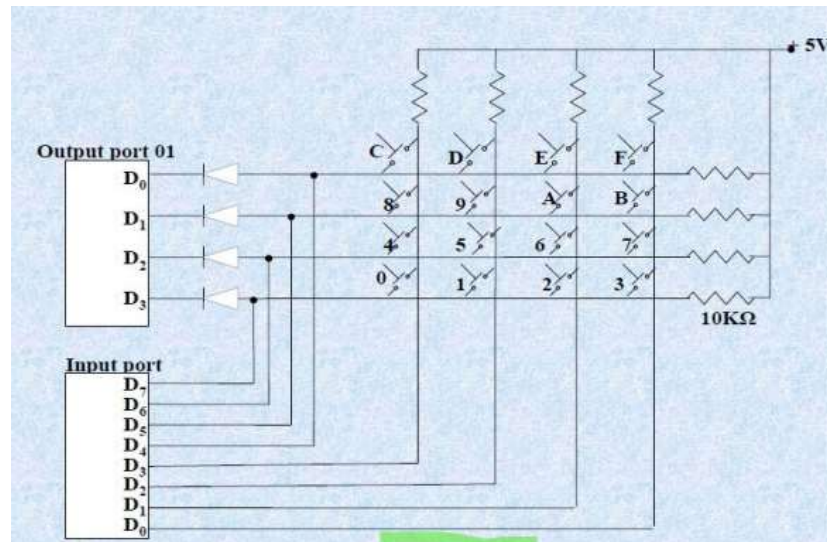
The RAM chips required = $2^{16} / 2^{12} = 2^4 = 16$

d) What will be the size of address decoder?

The size of the address decoder is 4 to 16.

B. Show the software keyboard interfacing process through a flowchart. Describe each step for 4x4 keyboard.





You need to broadly describe the procedure of how it's works.

C. Serialize the steps associated in turning on or off a pixel in LCD monitor.

Steps to Turn On or Off a Pixel in an LCD Monitor

- ✓ **Receive Signal:** The monitor gets a signal from the graphics card about which pixels to change.
- ✓ **Process Image:** It processes the signal to find out the color and brightness for each pixel.
- ✓ **Activate Controls:** The monitor activates the correct row and column for the pixel.
- ✓ **Apply Voltage:** A voltage is applied to the liquid crystals at that pixel.
- ✓ **Align Crystals:** The voltage changes how the liquid crystals align, allowing or blocking light.
- ✓ **Light Interaction:** If the pixel is "on," the backlight shines through; if "off," it stays dark.
- ✓ **Color Filtering:** The color filter allows specific colors to show through.
- ✓ **Update Display:** The pixel's state is updated, showing the correct color on the screen.
- ✓ **Refresh Display:** The monitor keeps refreshing the display to maintain the pixel changes.

A. Define software interrupt with example.

Interrupt generated by executing an instruction is called software interrupt. It's also called trap.

examples:

1. A cout (or) cin statement would generate a software interrupt because it would make a system call to print something.
2. A fork() statement in Linux would generate a software interrupt because it would make a system call to create a new process.

B. what are the problems of linear decoding system in main memory array design? How to solve them?

Problems of Linear decoding system in main memory array design:

1. Bus Conflict: two or more of lines A10-A13 are low at the same time, more than one SRAM chip are selected.

Solution: software must be written such that it never reads into or writes from any address in which more than one of bits A10-A13 are low.

2. Memory Foldback: Wastes a large amount of address space. For example, whenever the address value is B800 or 3800, SRAM chip-I is selected.

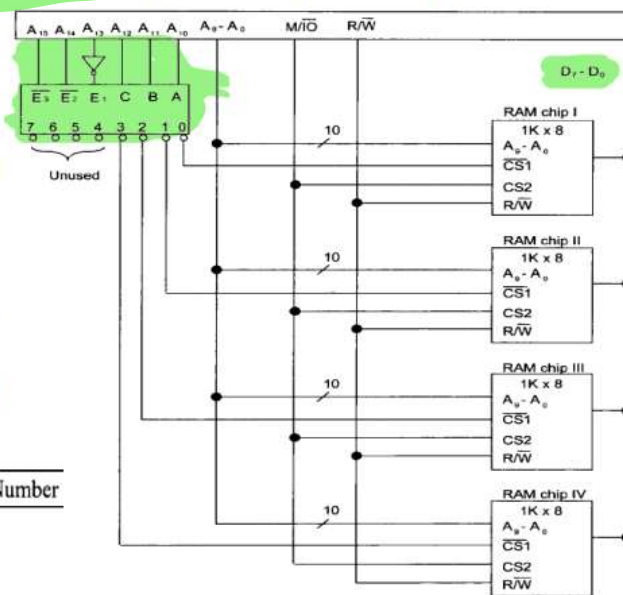
Solution: use full decoded memory addressing.

Main Memory Array Design

• **Full Decoding.**

- Using 3x8 decoder output selects one of the four 1K SRAM chips, depending on the values of A₁₂, A₁₁, and A₁₀.
- Decoder output enabled only when **E3 = E2 = 0 and E1 = 1**

A ₁₂	A ₁₁	A ₁₀	SRAM Chip Number
0	0	0	I
0	0	1	II
0	1	0	III
0	1	1	IV



Interconnecting a microprocessor with a 4K RAM using full decoded memory addressing.

C. Interfaces two chips of 4k RAM and 1 chip of 2k ROM with processor. Starting address of RAM is 8000H.

	A19	A18	A17	A16	A15	A14	A13	A12	A11	A10	A9	A8	A7	A6	A5	A4	A3	A2	A1	A0
8	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
8	F	F	F	1	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1
9	0	0	0	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
9	F	F	F	1	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1
A	0	0	0	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
A	F	F	F	1	0	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1
A	8	0	0	1	0	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0
A	7	F	F	1	0	1	0	0	1	1	1	1	1	1	1	1	1	1	1	1

Draw the block diagram from the above explanation.

1. A processor having 16 bits address line and it is byte addressable. Interface 4096x8 RAM with the processor.

1. What is the memory capacity of the processor?
2. How many addresses are required for a RAM?
3. How many RAM chips are required?
4. What will be the size of address decoder?

Already noted.

Session 2017-2018.

1.

A. Show the pin configuration of 8255 PPI and describe the operation performed by each unit separately.



Pin Diagram of 8255 PPI

Pin Configuration

- Pins 27 to 34: These pins are used for the data bus, which is tri-state and bidirectional. They connect the 8255 to the processor, allowing data and control signals to flow between them.
- Pin 26 (VCC): Provides the external voltage needed for the circuit.
- Pin 7: Serves as the ground connection for the circuit.
- Total Pins: Out of 40 pins, 24 are used for I/O ports, while the remaining pins are designated for control signals

B. Convert 39887.5625 to IEEE 32-bit floating point format.

- a. The integral part is $39887_{10} = 1001101111001111_2$. The fractional:
- | | | |
|---------------------------|---|--|
| $0.5625 \times 2 = 1.125$ | 1 | Generate 1 and continue with the rest. |
| $0.125 \times 2 = 0.25$ | 0 | Generate 0 and continue. |
| $0.25 \times 2 = 0.5$ | 0 | Generate 0 and continue. |
| $0.5 \times 2 = 1.0$ | 1 | Generate 1 and nothing remains. |
- So $39887.5625_{10} = 1001101111001111.1001_2$.

b. Normalize: $1001101111001111.1001_2 = 1.0011011110011111001_2 \times 2^{15}$.

c. Mantissa is 00110111100111110010000, exponent is $15 + 127 = 142 = 10001110_2$, sign bit is 0.
So 39887.5625 is 0 10001110 00110111100111110010000 $= 471bcf90_{16}$

C. Explain software keyboard interfacing procedure with circuit connections and algorithm.

2018-2019 session 3(B)

2.

A. Differentiate between programmed I/O and interrupt initiated I/O.

Aspect	Programmed I/O	Interrupt-Initiated I/O
Initiation	The CPU directly controls data transfer.	Data transfer is initiated by external hardware signals.
CPU Involvement	CPU actively polls or checks status to transfer data.	CPU is freed from continuously checking; waits for interrupt to handle data transfer.
Efficiency	Less efficient, as CPU needs to continuously check status, tying up resources.	More efficient, as CPU is freed for other tasks until interrupted.
Module	Its module is treated as a slow module.	Its module is faster than programmed I/O module.
Example	A simple example is reading data from a keyboard in which the CPU continuously checks if a key is pressed.	An example is reading data from a network card. The CPU can perform other tasks until the network card signals that data is ready to be read, triggering an interrupt.

B. How can the signed binary numbers be represented?

Signed binary numbers can be represented using the following methods:

- ✓ **Sign-Magnitude Representation:** The leftmost bit (most significant bit) indicates the sign (0 for positive, 1 for negative), while the remaining bits represent the magnitude.
- ✓ **1's Complement:** Negative numbers are represented by inverting all the bits of the corresponding positive number.
- ✓ **2's Complement:** Negative numbers are represented by inverting all the bits and then adding 1 to the least significant bit (LSB).

C. What is the range of signed values that may be stored in 20 bits?

For a 20-bit signed binary number, the range is:

Minimum value: $-2^{19} = -524,288$

Maximum value: $2^{19} - 1 = 524,287$

So, the range of signed values is from -524,288 to 524,287.

3.

A. How to connect a microprocessor to ROM/RAM chips? Illustrate and explain each technique.**B. When is an overflow flag set or reset? Give examples.**

An overflow flag is set or reset based on the outcome of arithmetic operations, particularly addition and subtraction. It indicates whether the result of an operation fits within the available storage space.

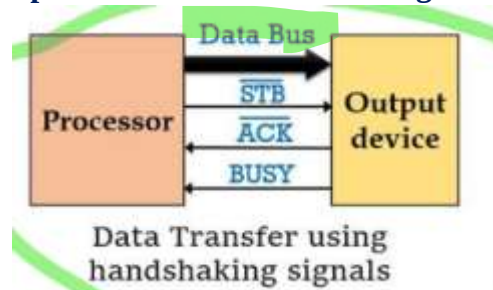
- **Set:** The overflow flag is set when the result of an operation exceeds the maximum value that can be represented in the available storage space.
 - **Example of Overflow Flag Set:** In an 8-bit system, if we add two numbers: $200 + 100$, the result is 300. However, since 300 cannot be represented using 8 bits (it's

larger than 255), the overflow flag would be set to indicate that the result is too large to fit in the available space.

- **Reset:** The overflow flag is reset when the result of an operation fits within the available storage space.

- Example of Overflow Flag Reset: In the same 8-bit system, if you add two numbers: $100 + 50$, the result is 150. Since 150 can be represented using 8 bits, the overflow flag would remain reset because the result fits within the available storage space

C. What are the basic principles of two-wire handshaking method of data transfer?



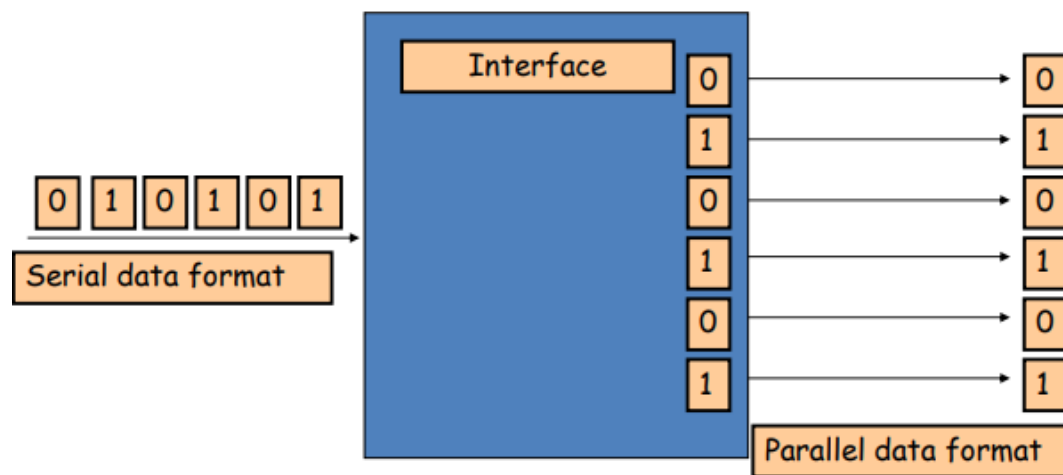
Principle of Handshaking

The basic principle of the two-wire handshaking method of data transfer is as follow:

One control line is in the same direction as the data flows in the bus from the source to destination. It is used by source unit to inform the destination unit whether there a valid data in the bus. The other control line is in the other direction from the destination to the source. It is used by the destination unit to inform the source whether it can accept the data. The sequence of control during the transfer depends on the unit that initiates the transfer.

1. A. What is serial and parallel communication? Explain with example.

- **Serial communication** transmits data one bit at a time, sequentially, making it simpler, cheaper, and more suitable for long distances.
Example: Sending data over a serial port (RS-232) connection between a computer and a printer. Each bit of the data (like a character or a command) is sent sequentially, one after the other, over a single wire.
- **Parallel communication** transmits data multiple bits at a time, simultaneously, making it faster, but more complex, expensive, and suitable for short distances.
Example: Sending data between a computer and its memory. In parallel communication, a group of wires is used to transfer multiple bits simultaneously, allowing for faster data transfer compared to serial communication.



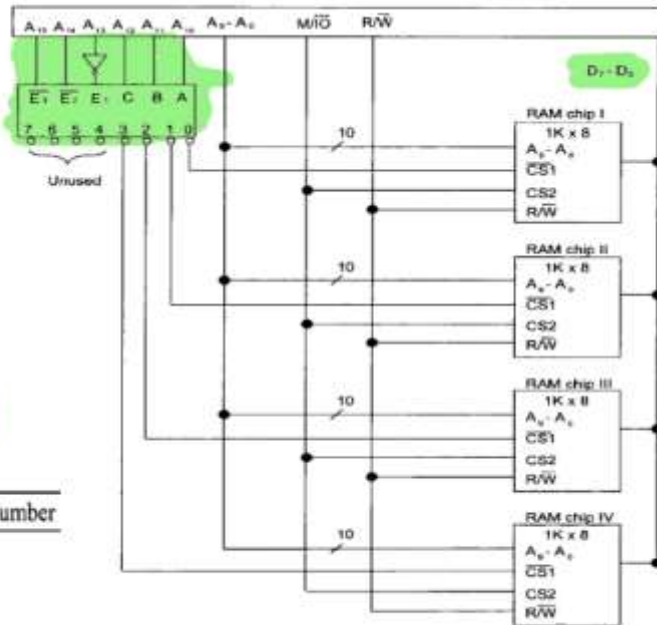
5.

- a) Illustrate an 8-bit microprocessor interfaced to a 4K RAM system using a full/partial decoded addressing technique. Also show the address map corresponding to the above mentioned organization.

• **Full Decoding.**

- Using 3x8 decoder output selects one of the four 1K SRAM chips, depending on the values of A_{12} , A_{11} , and A_{10}
- Decoder output enabled only when $E3 = E2 = 0$ and $E1 = 1$

A_{12}	A_{11}	A_{10}	SRAM Chip Number
0	0	0	I
0	0	1	II
0	1	0	III
0	1	1	IV



Interconnecting a microprocessor with a 4K RAM using full decoded memory addressing.

Binary Address Pattern	Device Selected	Address Assignment in Hex
$A_{15} A_{14} A_{13} A_{12} A_{11} A_{10} A_9 A_8 A_7 A_6 A_5 A_4 A_3 A_2 A_1 A_0$		
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	RAM CHIP 0	0000 to 03FF
0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0	RAM CHIP 1	0400 to 07FF
0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0	RAM CHIP 2	0800 to 0BFF
0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0	RAM CHIP 3	0C00 to 0FFF

b) What do you mean by ISA bus and PCI bus?

ISA Bus

- ✓ What it is: An old way to connect parts (like sound cards or network cards) to a computer.
- ✓ Data Width: Can handle 8 bits or 16 bits of data at once.
- ✓ Speed: Works at 8 MHz, which is slow by today's standards.
- ✓ Use: Mostly found in older computers.

PCI Bus

- ✓ What it is: A newer way to connect parts to a computer, replacing the ISA bus.
- ✓ Data Width: Can handle 32 bits or 64 bits of data at once.
- ✓ Speed: Works at 33 MHz to 66 MHz, which is faster than ISA.
- ✓ Use: Found in modern computers for connecting graphics cards and other devices.

c) Write down the steps involved in starting a printing sequence using a printer controller chip.

Steps to Start a Printing Sequence:

- ✓ **Initialize Printer:** Power on and configure the printer settings.
- ✓ **Send Data:** Transfer print data to the printer's buffer.
- ✓ **Start Print:** Issue "Start Print" command to begin printing.
- ✓ **Move Print Head:** Control print head and paper feed.
- ✓ **Apply Ink/Toner:** Activate ink or laser for printing.
- ✓ **Feed & Eject Paper:** Move paper through and eject after printing.
- ✓ **End Print:** Send "End Print" command and reset.
- ✓ **Check Errors:** Monitor and handle any errors (e.g., paper jam).

d) Explain the operation of a delta-delta shadow-mask CRT.

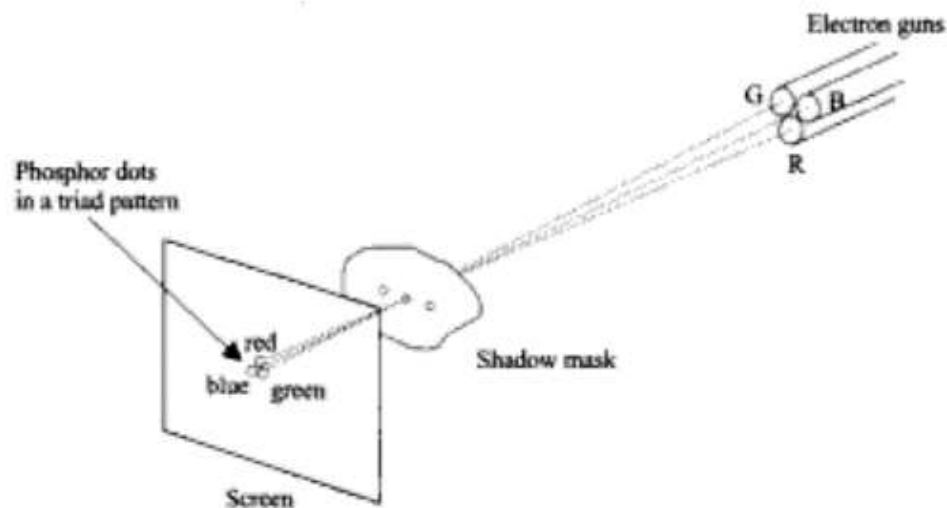


Fig. 2-6 Color CRT using a shadow mask.

Describe working principle and construction from the above diagram

6.

a) What are the basic differences between ASCII and EBCDIC?

Feature	ASCII	EBCDIC
Full Form	American Standard Code for Information Interchange	Extended Binary Coded Decimal Interchange Code
Bit Length	7-bit (128 characters) or 8-bit (256 characters)	8-bit (256 characters)
Character Set	Sequential and continuous	Non-sequential, with gaps between characters
Code for 'A'	65 (decimal)	193 (decimal)
Code for 'a'	97 (decimal)	129 (decimal)
Code for '0'	48 (decimal)	240 (decimal)
Primary Usage	General use in computers, programming, and the internet	IBM mainframes and legacy systems
Common File Types	Text files, HTML, programming languages	Legacy files on IBM systems

b) Describe the process of a device initiated I/O transfer.

Here are the steps of a device-initiated I/O transfer, described briefly:

1. **Device Request:** The I/O device needs to transfer data.
2. **Device Notifies CPU:** The device sends an interrupt signal to the CPU.
3. **Interrupt Handling:** The CPU temporarily stops its current task.
4. **Interrupt Service Routine (ISR):** The CPU runs a small program to handle the I/O request.
5. I/O Operation Initiation: The CPU sets up the parameters for the data transfer.
6. Data Transfer: Data is transferred between the device and memory.
7. Completion Notification: The device signals the CPU that the transfer is complete.
8. Interrupt Handling (Again): The CPU handles the completion interrupt.
9. Resume Execution: The CPU resumes its previous task

c) When is an overflow flag set or reset? Give examples?

Already noted

- d) Illustrate the character and signal generation procedure of a CRT display with a short description.

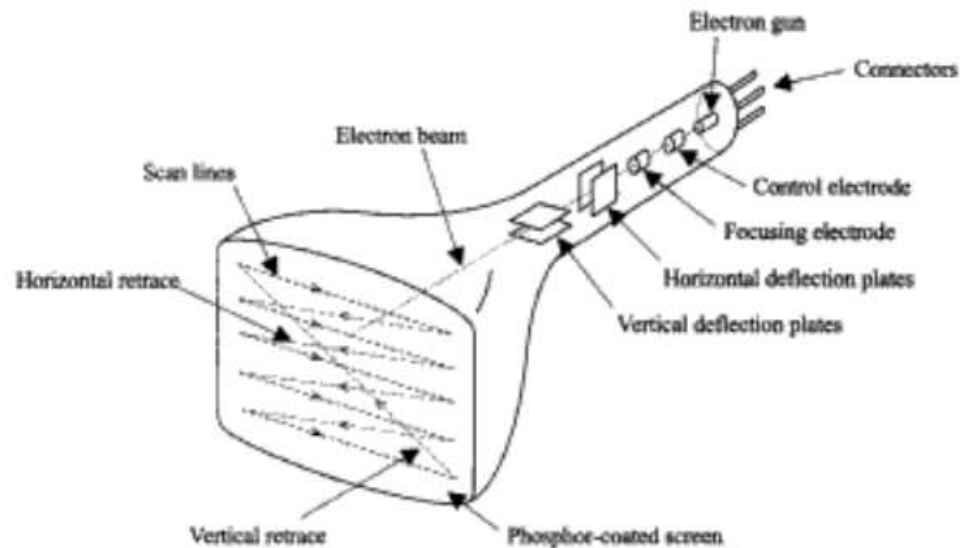


Fig. 2-5 Anatomy of a monochromatic CRT.

Describe working principle and construction from the above diagram.

- e) **The MMU can perform address translation in one of the two ways. Define them.**

The MMU can perform address translation in two main ways:

1. **Segmentation:**

- ✓ Divides memory into segments (e.g., code, data, stack), each with a base address and size.
- ✓ Logical addresses have a segment number and offset.
- ✓ Useful for organizing memory logically and providing protection, but can lead to fragmentation.

2. **Paging:**

- ✓ Divides memory into fixed-size pages and maps them to physical page frames.
- ✓ Logical addresses have a page number and offset.
- ✓ Simplifies memory management and avoids fragmentation but requires extra memory for page tables.

Session 2012-2013.

1. A) What is buffer? Why we use buffers?

A buffer is an area of memory used for the transfer of data between a computer and a peripheral.

- A buffer provides temporary storage of data.
- Using a buffer provides a link between a device and the processor and helps compensate for any differences in their working speeds.

B) Define interfaces. What are the characteristics of computer peripherals.

The interface is the combination of hardware and software needed to link the CPU to the peripherals and to enable them to communicate with the CPU despite all their differing characteristics.

Computer peripherals have different characteristics.

For example, they may:

- Have different data transfer rates;
- Use a wide variety of codes and control signals;
- Transmit data in serial or in parallel form;
- Even work at higher voltages than the CPU; &
- All operate at much slower speeds than the CPU.

C) Differentiate between buffers and spoolers.

Already noted.

D) What do you mean by solid state storage device? Write down the features and disadvantages of solid state storage devices.

A solid state storage device is a type of storage device that uses flash memory to store data. Unlike traditional hard drives, it has no moving parts, which makes it faster and more durable.

Features:

- ✓ Speed: Faster data access and transfer speeds.
- ✓ Durability: More resistant to physical shock and damage.
- ✓ Quiet Operation: No moving parts, so they are silent.
- ✓ Low Power Consumption: Uses less power than traditional hard drives.
- ✓ Compact Size: Smaller and lighter than hard drives.

Disadvantages:

- ✓ Cost: Generally, more expensive per gigabyte than traditional hard drives.
- ✓ Limited Write Cycles: Can wear out over time with many write and erase cycles.
- ✓ Capacity: Typically offers less storage capacity compared to the highest-capacity hard drives.